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Throughout this report, we are just going to go through how we proceeded throughout this project over the summer and what it eventually led to. In hopes that future years taking on this project can use this documentation to add to the project a solid foundation of understanding of where we finished.

The first meeting was pretty simple; we didn't have anything, so we ordered some parts we needed, which were an ESP32 micro board with a camera, buck converters, and necessary wiring. The idea was to just get the necessary parts, which we did, and then, from the beginning, software and assembly. All the presentations for the week-to-week will be posted as slideshows in GitHub.

Second week was almost a wash; all we started doing was using the software to start identifying objects using the computer camera. The parts didn't come in yet, but were ordered on this date. We also learned that taking away the remote control would be an easy process once the parts came in.

The third week we began wiring, and the parts worked correctly. At this point, we needed to 3D print mounts that could secure these parts. The buck converters, which take high voltage from the car and convert it to low voltage, were easy to assemble onto the car. The board and camera, on the other hand, took time to create a design that would fit in the spot we had envisioned for it. After 3 different printing attempts, we failed to create a mount for the camera and board this week.

Week 4, we began by successfully printing the first iteration of the ESP32 board and camera mount. Once we accomplished this, we were able to transfer driving procedures to the computer using the WASD keys. This was a good starting point for achieving full autonomy. However, around this point in time, we realized that the camera was extremely laggy, and other forms of sensing methods would need to be utilized in order for there to be minimal problems in obstacle detection.

Week 5, we made a purchase for a LiDAR sensor, which would work extremely well in detecting the distances of obstacles in front of the car. The new sensor would use a sort of echolocation, in a sense, to understand how far obstacles were. This week we ordered this part, worked on the mount for it before it even came, and also began coding software so that when it came, it would be ready to go.

Week 6, the LiDAR sensor came; we needed to reprint the mount for it and attach it to the front, which thankfully worked the first time around. Then assembled, we realized how good a piece of equipment this sensor actually was. It was able to map out the room really well. After that, we began training the AI by taking loads and loads of pictures of desired obstacles like

trees, people, traffic cones, and bushes. This was an extremely tedious part of the project, yet we did get 500 images and felt like the AI was at least decently trained for image recognition from the camera.

Week 7, we spent more time training the AI, and it did take an obscene amount of time; however, once this part was done, it was pretty smooth sailing.

Week 8, we felt really good and went to Stevens Park to test it. Obstacle avoidance was great in the lab; however, once we went to the park, the camera was lagging excessively. The LiDAR worked fine, but the throttle also stopped working. This was bad news, and we realized the board was too small to handle the LiDAR and camera input while also controlling the steering and throttle of the car. With that being said, everything eventually crashed, and we had to get new parts ASAP. We decided to get a larger board with a separate camera and new buck converters.

Week 9, we spent most of the time creating new mounts for all of the parts. This took up our whole day; however, once we had everything secure, we were able to very basically test the program, and everything was working well.

Week 10, we focused on getting it effectively working in Stevens Park. We were out there testing for about an hour and eventually got a good trial run we felt confident about showing during the presentation. We finalized a lot of documentation on top of that as well. It's important to note that the final product has really rough obstacle avoidance and could still implement better parts for smoother avoidance. The frame of the car could be better to help in more unpredictable terrain, as the small RC car frame limited where it was able to drive. Further training of the AI to allow it to recognize tree roots would also be beneficial. The first year of the project produced a rough working design that could certainly be improved upon if it were to be carried out for numerous years after this one.